

ST01

Physics in Everyday Life

Science and Technology carries 18-23 questions in every RAS paper and physics supplies 5-7 of them. Units, light and electricity are the three reliable suppliers. Device-to-principle matching and unit-to-quantity matching come up almost every year - build those two tables and you have banked most of the chapter.

Every physics question RPSC asks is really a question about something you have already seen - a pressure cooker whistling, a fuse blowing, a rear-view mirror, a blue sky. The paper never asks you to calculate. It asks you to name the principle behind the everyday thing. So learn it that way: phenomenon first, principle second.

Physics in Everyday Life

Units

- 7 SI base units
- newton, pascal, joule, watt
- Light year, parsec =
- DISTANCE**
- Dioptre = lens power
- 1 hp = 746 W

Fluids

- Pascal - hydraulic brake
- Archimedes - ship, lactometer
- Bernoulli - aerofoil, atomiser
- Surface tension - raindrop

Light (biggest)

- TIR - mirage, diamond, fibre
- Scattering - blue sky, red sunset
- Myopia concave, hypermetropia convex
- Dispersion - rainbow, VIBGYOR

Motion

- 1st = inertia (seat belt)
- 2nd = $F = ma$
- 3rd = rocket, gun recoil
- Impulse - airbag, sand pit

Heat

- Water sp. heat 4.2 J/g/K
- Latent heat - steam burns worse
- Ice floats - density max at 4°C
- Thermos blocks all three modes

Electricity

- $V = IR$, 220 V 50 Hz AC
- Fuse in SERIES in LIVE wire
- Transformer - mutual induction, AC only
- Motor left hand, generator right hand

Gravitation

- g max at poles, zero at centre
- Escape velocity 11.2 km/s
- Geostationary 36,000 km
- Weightlessness = free fall

Sound

- No sound in vacuum
- 20 Hz - 20 kHz audible
- Ultrasound - SONAR, sonography
- Echo needs 17.2 m

Modern physics

- Fission - reactor, U-235
- Fusion - the Sun
- Gamma most penetrating
- Roentgen X-rays, Becquerel radioactivity

Units and measurement - the easiest marks in the paper

Measurement means comparing a quantity with an agreed standard. That standard is the unit. The SI system fixes seven of these as base units and builds everything else out of them. RPSC rarely asks what a unit IS. It asks which quantity a unit belongs to - and it deliberately picks units that sound like one thing and measure another. Learn the seven, learn the common derived ones, then spend most of your time on the trap list.

The seven SI base units - memorise this whole grid

Base unit	Symbol	Quantity measured
metre	m	Length
kilogram	kg	Mass
second	s	Time
ampere	A	Electric current
kelvin	K	Thermodynamic temperature
mole	mol	Amount of substance
candela	cd	Luminous intensity

Derived units you must be able to match both ways

Unit	Quantity	In base terms / note
newton (N)	Force	kg m/s ²
pascal (Pa)	Pressure, stress	N/m ²
joule (J)	Work, energy, heat	N m
watt (W)	Power	J/s
hertz (Hz)	Frequency	per second
coulomb (C)	Electric charge	A s
volt (V)	Potential difference	J/C
ohm	Resistance	V/A
farad (F)	Capacitance	C/V
tesla (T), weber (Wb)	Magnetic flux density, magnetic flux	-
henry (H)	Inductance	-

Unit	Quantity	In base terms / note
diopetre (D)	Power of a lens	1 / focal length in metres

- Pressure units: pascal, bar, torr, atmosphere, mm of mercury - all of these names are pressure.
- $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} = 760 \text{ torr} = 76 \text{ cm of mercury} = 1.013 \text{ bar}$.
- $1 \text{ horsepower} = 746 \text{ watt}$. 1 kilowatt-hour (the Board of Trade unit on your electricity bill) = $3.6 \times 10^6 \text{ joule}$.
- Small lengths: angstrom = 10^{-10} m ; nanometre = 10^{-9} m ; micron = 10^{-6} m ; fermi = 10^{-15} m .
- Scalars have magnitude only - mass, speed, distance, work, energy, power, temperature, density, pressure.
- Vectors have magnitude and direction - displacement, velocity, acceleration, force, momentum, impulse, torque, weight.

TRAP

The unit trap RPSC loves: light year and parsec measure DISTANCE, not time (1 parsec = 3.26 light years). Barn = area. Knot = speed. Decibel = sound level. Diopetre = power of a lens. Poise = viscosity. Chandrasekhar limit = a MASS. And weight is a force, so its unit is the newton, not the kilogram.

ASKED IN RAS

Asked in RAS Pre 2024 - out of bar, pascal, torr and atmosphere, which are units of pressure? Answer: all four. The examiner was betting you would think 'bar' is something else.

Motion, Newton's laws and momentum - explained through the bus you travel in

Newton's three laws are not abstract. Each one describes something that happens to you daily. The first says a body keeps doing whatever it was doing unless a force interferes - that is why you lurch forward when the bus brakes. The second connects force to the change it produces: $F = ma$, or more properly, force is the rate of change of momentum. The third says forces always come in pairs, equal and opposite, acting on two different bodies.

Law to everyday example - this is how the matching question is set

Law	What it says	Everyday proof
First law (law of inertia)	A body stays at rest or in uniform motion unless an external force acts	Passenger lurching in a braking bus; seat belts; dust flying out when a carpet is beaten; a coin staying put when the card under it is flicked
Second law	Force = mass x acceleration = rate of change of momentum	A cricketer pulls his hands back while catching; heavier vehicles need bigger brakes
Third law	Every action has an equal and opposite reaction	Rocket and jet propulsion; recoil of a gun; walking; swimming; a boat moving back when you step off it

- *Momentum = mass x velocity, and it is a vector. Conservation of linear momentum explains gun recoil and rocket propulsion.*
- *Impulse = force x time = change in momentum. The same change in momentum spread over more time means a smaller force.*
- *That is exactly why air bags, sand pits at a long jump, foam packing and a cricketer's follow-through reduce injury.*
- *Inertia depends only on mass - a heavier body has more inertia. Inertia of rest, of motion and of direction.*
- *Centripetal force keeps a body on a circular path and points towards the centre; centrifugal force is the apparent outward pull felt by the rotating observer, not a real force.*
- *Work is done only when a force produces displacement along its own direction: work = force x displacement x cosine of the angle.*
- *A coolie carrying a load on his head across a level platform does ZERO work against gravity - the force is vertical, the motion horizontal.*
- *Energy is the capacity to do work; it is never destroyed, only converted. Power is work done per unit time, measured in watt.*
- *Trick to remember the third law: ROCKET, RIFLE, ROWING, RUNNING. All four R's are action-reaction.*

Gravitation, satellites and why astronauts float

Every two bodies in the universe attract each other - that is Newton's universal law of gravitation. On Earth that attraction gives every falling body the same acceleration, g . Mass is how much matter you contain and never changes. Weight is the gravitational force on that mass, so weight changes with place. Get this one distinction right and half the gravitation questions solve themselves.

- $g = 9.8 \text{ m/s}^2$ at sea level. It is **MAXIMUM** at the poles and **MINIMUM** at the equator, because the Earth bulges at the equator and spins.
- g decreases as you go up from the surface, and also as you go down into the Earth - it becomes **ZERO** at the centre of the Earth.
- g on the Moon is about one-sixth of g on Earth. A 60 kg man still has 60 kg of mass on the Moon, but weighs one-sixth as much.
- Escape velocity of the Earth = 11.2 km/s; of the Moon = 2.38 km/s. The Moon's escape velocity is so low that gas molecules leak away - that is why the Moon has no atmosphere.
- Orbital velocity of a near-Earth satellite is about 7.9 km/s. Escape velocity does not depend on the mass or the direction of the body projected.
- Repeatedly set as a statement question: ' g is zero at the centre of the Earth' is **TRUE**; 'escape velocity depends on the mass of the body' is **FALSE**.

Satellite orbits - match the orbit to the job

Orbit	Height and period	What it is used for
Geostationary (geosynchronous, equatorial, circular)	About 36,000 km (35,786 km); period 24 hours	Communication, television broadcasting, weather watching - it appears fixed over one point
Polar / sun-synchronous	About 800-900 km	Remote sensing, resource survey, mapping - it crosses each place at the same local time

TRAP

Weightlessness in an orbiting spacecraft is **NOT** because gravity has vanished. Gravity is very much there - it is what holds the craft in orbit. The astronaut floats because the astronaut and the spacecraft are falling together at the same rate. The same weightlessness is felt inside a lift whose cable snaps.

Pressure and fluids – Pascal, Archimedes, Bernoulli and the

pressure cooker

Pressure is force spread over area. Same force, smaller area, more pressure - which is why a sharp knife cuts and a blunt one does not, why a camel's broad feet do not sink into sand, and why school bags are given wide straps. Fluids, meaning liquids and gases, carry pressure in three ways that RPSC asks about constantly. Three names, three sets of devices. Learn them as one block.

The three fluid principles and their devices – the highest-yield table in this section

Principle	In one plain sentence	Devices and effects
Pascal's law	Pressure applied to an enclosed fluid is transmitted equally in all directions	Hydraulic press, hydraulic brakes of a car, hydraulic jack, hydraulic lift
Archimedes' principle	A body immersed in a fluid is pushed up by a force equal to the weight of the fluid it displaces	Ship, submarine, hydrometer, lactometer (purity of milk); a floating body displaces its own weight
Bernoulli's principle	Where a fluid flows faster, its pressure is lower	Lift on an aeroplane wing (aerofoil), atomiser and spray gun, chimney draught, two boats sailing side by side drawn together, a roof blown off in a storm

- Atmospheric pressure at sea level = 76 cm of mercury = 1.013×10^5 pascal. It is measured by a barometer; Torricelli made the first one.
 - Atmospheric pressure falls with altitude. Lower pressure means a lower boiling point, so water boils below 100 C on a hill and food cooks slowly.
 - A pressure cooker raises the pressure inside, which raises the boiling point of water above 100 C - the food cooks faster because it cooks HOTTER, not because steam is trapped.
 - Nose bleeding at high altitude, breathlessness on mountains and a fountain pen leaking in an aircraft are all low-pressure effects.
 - A sudden fall of the barometer reading warns of a storm; a slow fall means rain; a rise means dry weather.
-
- Surface tension makes a liquid surface behave like a stretched skin. It gives raindrops their spherical shape, lets a greased needle float, and lets insects walk on water.
 - Soap and detergent LOWER the surface tension of water, which is why soapy water spreads into cloth and cleans. Surface tension also falls as temperature rises - hot water washes better.
 - Capillarity is the rise of liquid in a fine tube: oil rises in a lamp wick, ink spreads on blotting paper, water rises in soil and in plant stems.
 - Mercury does NOT wet glass, so mercury is depressed in a capillary tube instead of rising in it.
 - Viscosity is a fluid's internal friction. Viscosity of a LIQUID decreases with rise in temperature; viscosity of a GAS increases with temperature. Units: poise (CGS), pascal-second (SI).

ASKED IN RAS

A documented repeat - the working of a hydraulic brake is based on Pascal's law, and 'food cooks faster in a pressure cooker because the boiling point of water rises with pressure'. Both have been set more than once. Do not answer 'because steam carries more heat'.

Heat – specific heat, latent heat and the three ways heat travels

Heat is energy in transit from a hotter body to a colder one; temperature only tells you which way it will flow. Two numbers control almost every heat question. Specific heat is how much heat a substance needs to warm up by one degree. Latent heat is the heat a substance swallows to change its state with no change in temperature at all. Water is exceptional in both, and the exam lives on that exception.

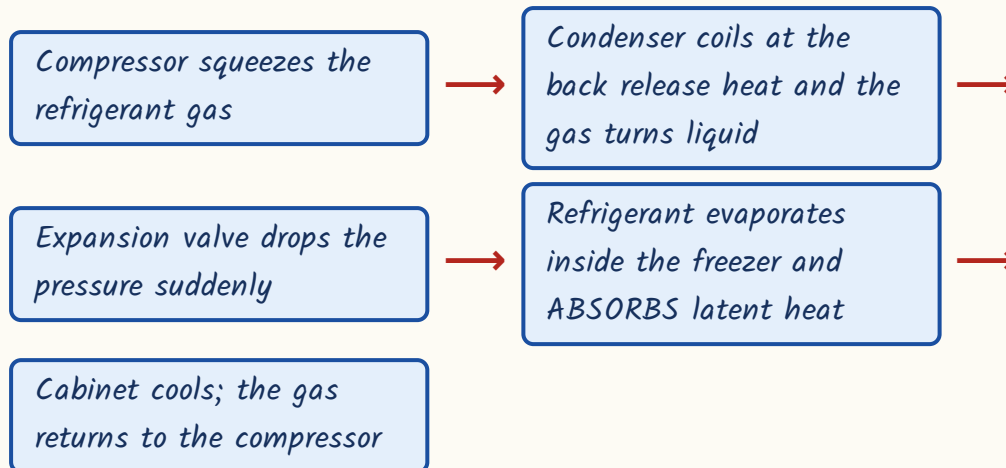
- Specific heat of water = 4.2 J/g/K , the highest among common liquids. That is why water is the coolant in car radiators and why the sea moderates coastal climate.
- Latent heat of fusion of ice = 80 cal/g (336 J/g). Latent heat of vaporisation of water = 540 cal/g (2260 J/g).
- Steam at 100°C burns far worse than boiling water at 100°C because it must first give up that 540 cal/g of latent heat on your skin.
- Anomalous expansion of water: between 0°C and 4°C water **CONTRACTS** on heating, so its density is maximum at 4°C .
- That is why ice floats and a lake freezes from the surface downwards, leaving liquid water below in which fish survive the winter.
- $0 \text{ kelvin} = -273.15^\circ \text{C}$. Minus 40 degrees is the same number on the Celsius and the Fahrenheit scale.
- Evaporation always cools - sweating cools the body, water stays cool in an earthen matka, and the refrigerant cools the fridge, all by absorbing latent heat.
- A thermos flask blocks all three modes at once: the vacuum stops conduction and convection, the silvered walls stop radiation, the cork stopper reduces conduction.
- A bimetallic strip bends because two metals expand unequally - it is the thermostat inside an electric iron, a geyser and a fire alarm.
- Greenhouse effect: short-wavelength sunlight passes in, but the long-wavelength infrared re-radiated by the ground is trapped by carbon dioxide, water vapour and methane.
- Adding salt raises the boiling point of water and lowers its freezing point - which is why salt is spread on icy roads.

The three modes of heat transfer - if the question says vacuum, the answer is radiation

Mode	Needs a medium?	Everyday example
Conduction	Yes - and no bulk movement of matter; chiefly in solids	A metal spoon heating up in hot tea; silver is the best conductor of heat, then copper

Mode	Needs a medium?	Everyday example
Convection	Yes - the fluid itself moves	Land breeze and sea breeze; boiling water; a room heater warming the air; the chimney
Radiation	NO medium needed	Heat from the Sun crossing empty space; a bonfire warming your face

How a refrigerator cools - trace the refrigerant



Sound - why there is silence in space and how SONAR finds the sea floor

Sound is a mechanical wave. It needs particles to pass the vibration along, so it cannot cross a vacuum - there is no sound in space, and an electric bell ringing inside an evacuated jar goes silent. Because the particles of a solid sit closest together, sound travels fastest in solids and slowest in gases.

- Speed of sound in air is about 332 m/s at 0 °C. It increases with temperature and humidity; it does not depend on pressure.
- Order of speed: solids > liquids > gases. Sound travels roughly four times faster in water than in air.
- The audible range for humans is 20 Hz to 20,000 Hz. Above 20,000 Hz is ULTRASONIC; below 20 Hz is INFRASONIC.
- Loudness depends on amplitude (measured in decibel), pitch on frequency, and quality or timbre on the waveform - timbre is how you recognise a voice on the phone.
- An echo is heard only if the reflector is at least about 17.2 m away, because the ear holds a sound for about 0.1 second.
- A stethoscope works by MULTIPLE REFLECTION of sound inside its tube. Sound boards in an auditorium and a megaphone also work by reflection.
- SONAR = Sound Navigation and Ranging. It sends an ultrasonic pulse down and times the echo to find sea depth, shoals of fish, submarines and wrecks.
- Medical uses of ultrasound: sonography of the abdomen and foetus, echocardiography of the heart, and lithotripsy to break kidney stones without surgery.
- Industrial uses of ultrasound: detecting cracks and flaws in metal castings, and cleaning parts a brush cannot reach.
- Bats and dolphins navigate by echolocation. Elephants and whales communicate using infrasound.
- Doppler effect - the apparent change in frequency when source and observer move relative to each other: the rising then falling whistle of a passing train, radar speed guns, and the red shift of receding galaxies.
- Noise limit for a residential zone under India's Noise Pollution (Regulation and Control) Rules, 2000: 55 dB(A) by day and 45 dB(A) by night.

Sound by frequency - the three bands

ULTRASONIC

SONAR, sonography, lithotripsy; bats

above 20,000 Hz

AUDIBLE to humans

loudness in decibel, pitch from frequency

20 Hz to 20,000 Hz

INFRASONIC

elephants and whales communicate here

below 20 Hz

ASKED IN RAS

Asked in RAS Pre 2021 - a stethoscope works on the principle of MULTIPLE REFLECTION of sound. Not refraction, not resonance. The ultrasound and SONAR statement set is a documented repeat cluster as well.

Light and optics - the heaviest scoring section in the whole chapter

Light travels at 3×10^8 m/s in vacuum. Everything you will be asked comes out of four behaviours - it bounces (reflection), it bends when it changes medium (refraction), it splits into colours (dispersion), and it gets thrown sideways by tiny particles (scattering). Learn which everyday sight belongs to which behaviour. That is the entire section.

- Laws of reflection: the angle of incidence equals the angle of reflection, and the incident ray, the reflected ray and the normal all lie in one plane.
- A plane mirror gives a virtual, erect image of the same size, as far behind the mirror as the object is in front, and laterally inverted - which is why AMBULANCE is painted in mirror writing.
- A concave mirror is converging; a convex mirror is diverging and always gives an erect, diminished image with a wide field of view.
- Refraction: light entering a denser medium bends TOWARDS the normal and slows down. Refractive index = speed of light in vacuum divided by speed in the medium.
- A stick looks bent in water, a coin in a bowl seems to rise, and print looks raised under a glass slab - all plain refraction.

Mirror or lens - match the device to the optic used

Device	Optic used	Why
Rear-view mirror of a vehicle	Convex mirror	Always erect and diminished, gives a wide field of view
Vehicle headlight, torch, searchlight	Concave mirror	Source at the focus throws out a parallel beam
Shaving mirror, dentist's mirror	Concave mirror	Gives an enlarged erect image when the object is inside the focus
Solar cooker and solar furnace	Concave mirror	Concentrates sunlight at the focus
Simple magnifying glass	Convex lens	Enlarged virtual image
Compound microscope	Two convex lenses	Objective plus eyepiece
Astronomical telescope	Convex objective of long focal length + convex eyepiece; a reflecting telescope uses a concave mirror instead	Gathers light from distant objects
Periscope	Plane mirrors, or total-reflecting prisms	Bends the line of sight twice

Device	Optic used	Why
Endoscope	Optical fibres	Carries light around bends inside the body

Now total internal reflection - the single most asked idea in optics. When light tries to go from a denser medium into a rarer one at an angle bigger than the critical angle, it does not escape at all. It is reflected wholly back inside.

Diamond has a very small critical angle, about 24 degrees, so light entering it bounces around many times before it can get out. That is the sparkle.

- Total internal reflection explains the mirage on a hot road, the sparkle of a cut diamond, the shining of a crack in glass, and the working of an optical fibre.
- An optical fibre has a core of higher refractive index and a cladding of lower refractive index - light is trapped in the core by repeated total internal reflection.
- Optical fibre uses: telecommunication and broadband, endoscopy, decorative lamps, and signal transmission immune to electrical interference.
- Atmospheric refraction explains the twinkling of stars, the Sun being seen about two minutes before actual sunrise and after actual sunset, and the oval look of the Sun at the horizon.
- Planets do not twinkle because they are near enough to be extended sources - the flickers from their different points average out.

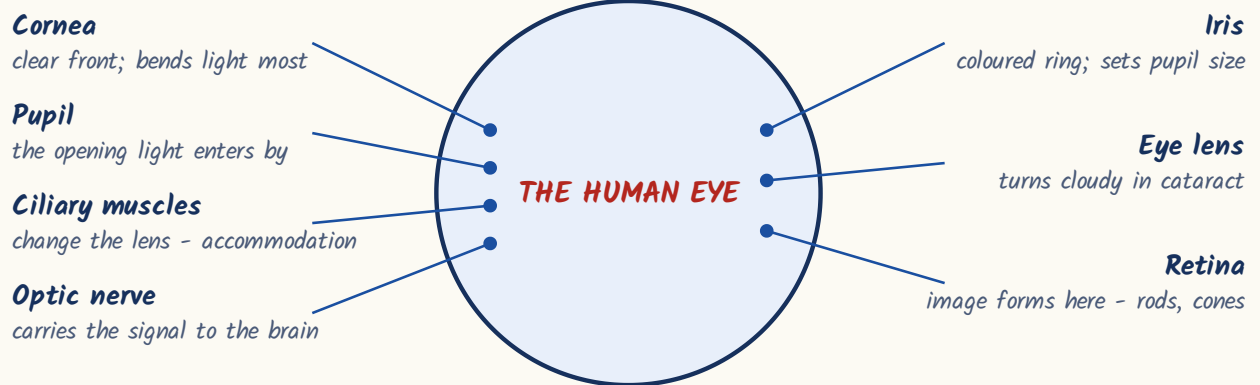
Defects of vision and their corrective lenses - learn this cold

Defect	What goes wrong	Corrective lens
Myopia (short-sightedness)	Image forms IN FRONT of the retina; distant objects blurred	Concave (diverging) lens
Hypermetropia (long-sightedness)	Image forms BEHIND the retina; near objects blurred	Convex (converging) lens

Defect	What goes wrong	Corrective lens
Presbyopia	Old age - the lens stiffens and loses accommodation, both near and far affected	Bifocal lens
Astigmatism	Uneven curvature of the cornea; lines in one direction blurred	Cylindrical lens
Cataract	The eye lens itself turns cloudy owing to protein changes	No lens helps - corrected by surgery

- Power of a lens in dioptre = 1 divided by the focal length in metres. A convex lens has POSITIVE power, a concave lens NEGATIVE power.
- The normal adult eye focuses comfortably from 25 cm - the least distance of distinct vision - out to infinity. The ciliary muscles change the lens curvature; that is accommodation.
- The retina carries rods for dim light and cones for colour and bright light; colour blindness is a defect of the cones.
- Dispersion: a prism splits white light into VIBGYOR because each colour bends by a different amount. Violet bends MOST, red bends LEAST.
- A rainbow is refraction plus dispersion plus internal reflection inside raindrops. It always appears in the part of the sky opposite the Sun.
- LASER stands for Light Amplification by Stimulated Emission of Radiation - its light is monochromatic, coherent and highly directional.

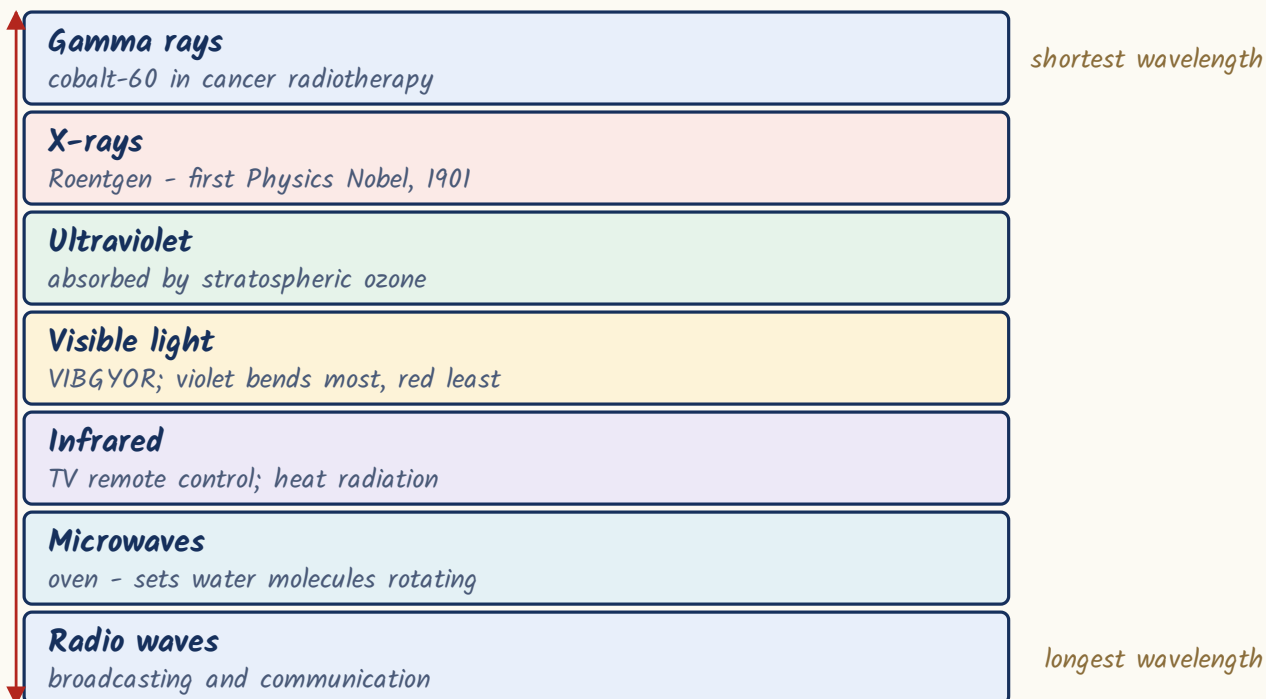
The human eye - the parts that are asked



Scattering closes the section and it is the favourite question of the lot. When sunlight meets molecules much smaller than its wavelength, the shorter wavelengths are scattered far more than the longer ones. Blue is scattered most, red least. So the sky you see is scattered blue light. At sunrise and sunset the light travels a long slanting path through the atmosphere, the blue is scattered away entirely, and only red and orange survive to reach your eye.

- Blue sky, red sunrise and sunset, and the reddish look of the Sun at the horizon - all scattering.
- The sky looks BLACK to an astronaut and on the Moon because there is no atmosphere to scatter light.
- Clouds look white because their water droplets are large and scatter all wavelengths about equally.
- Danger signals and traffic stop lights are RED because red scatters the least and travels furthest through fog and dust.
- Electromagnetic spectrum in increasing order of wavelength: gamma rays, X-rays, ultraviolet, visible light, infrared, microwaves, radio waves. The order of frequency is exactly the reverse.
- A television remote control uses INFRARED. A microwave oven uses microwaves, which heat food by making its water molecules rotate.

Electromagnetic spectrum - wavelength grows downwards



ASKED IN RAS

Asked in RAS Pre 2023 - match the defect of vision to the corrective lens (myopia to concave, hypermetropia to convex, presbyopia to bifocal, astigmatism to cylindrical). Asked in RAS Pre 2018 - the waves used in a TV remote are infrared. Blue sky, optical fibre and total internal reflection are all documented repeats.

YAAD RAKHO

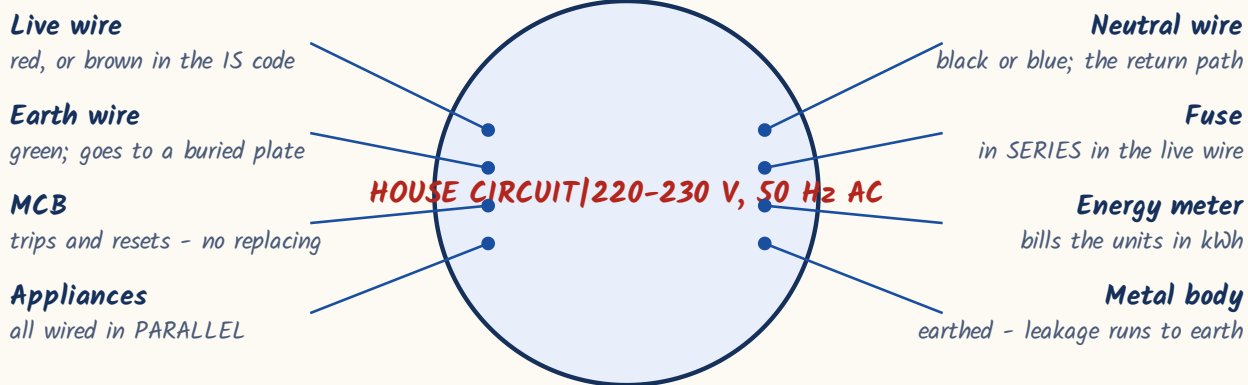
Remember: MY-opia is MY-nus - the concave lens has negative power. And for scattering, 'short scatters' - the short wavelength (blue) scatters most, so the sky is blue and the sunset, having lost its blue, is red.

Electricity and magnetism – the fuse, the earth wire and the transformer

Ohm's law is the backbone: at constant temperature the current through a conductor is proportional to the potential difference across it, $V = IR$. Everything in your house is built on top of that one relation plus two safety ideas – a deliberately weak link (the fuse) and a deliberate escape route to the ground (the earth wire).

- Indian domestic supply is alternating current at about 220-230 volt and 50 hertz.
- Direct current flows one way and comes from cells and batteries; alternating current reverses direction fifty times a second and is used for supply because only AC can be stepped up and down by a transformer.
- Household appliances are wired in PARALLEL, so each gets the full voltage and each can be switched independently.
- An ammeter is connected in SERIES and has very low resistance; a voltmeter is connected in PARALLEL and has very high resistance.
- Power = voltage x current. Electrical energy is billed in kilowatt-hours; 1 kWh = 3.6×10^6 joule.
- The resistance of a metal conductor INCREASES with temperature; the resistance of a semiconductor DECREASES with temperature.
- The heating effect of current runs the electric iron, heater, geyser and filament bulb; nichrome is used for heating elements because of its high resistance and high melting point.
- Earthing joins the metal body of an appliance to a thick wire buried in the earth. If the live wire touches the body, the leakage current runs safely to earth instead of through you.
- Traditional Indian wiring colours: red = live, black = neutral, green = earth. The newer IS colour code uses brown for live, light blue for neutral and green-yellow for earth.
- Short circuit - live and neutral touch directly, resistance collapses and current shoots up. Overloading - too many appliances drawing from one line.
- A transformer works on MUTUAL INDUCTION and works only with alternating current - it will do nothing with DC.
- A step-up transformer raises voltage and lowers current; a step-down transformer does the reverse. High-voltage transmission is used because it cuts the heat loss in the wires.

The domestic wiring circuit



Fuse vs MCB - the exact contrast RPSC sets

Electric fuse	MCB (miniature circuit breaker)
A thin wire of a tin-lead alloy with a low melting point	A switch containing a bimetallic strip plus an electromagnet
Connected in SERIES, in the LIVE wire	Also in the live line, mounted in the distribution board
Melts and breaks the circuit on overload or short circuit	Trips and springs open on overload or short circuit
Must be replaced once it has blown	Simply reset by pushing the lever back up
Slower to act, but cheap	Faster and more precise, and reusable

Device to energy conversion - a standing matching question

Device	Converts	Principle
Electric motor	Electrical energy into mechanical energy	Force on a current-carrying conductor in a magnetic field; Fleming's LEFT-hand rule
Generator or dynamo	Mechanical energy into electrical energy	Faraday's electromagnetic induction; Fleming's RIGHT-hand rule; Lenz's law gives the direction
Photovoltaic (solar) cell	Solar energy DIRECTLY into electrical energy	Photoelectric behaviour of a semiconductor junction

Device	Converts	Principle
Microphone	Sound into electrical energy	-
Loudspeaker	Electrical energy into sound	-
Electric bulb / LED	Electrical energy into light and heat	Heating effect of current; an LED is a semiconductor diode
Battery or dry cell	Chemical energy into electrical energy	-

ASKED IN RAS

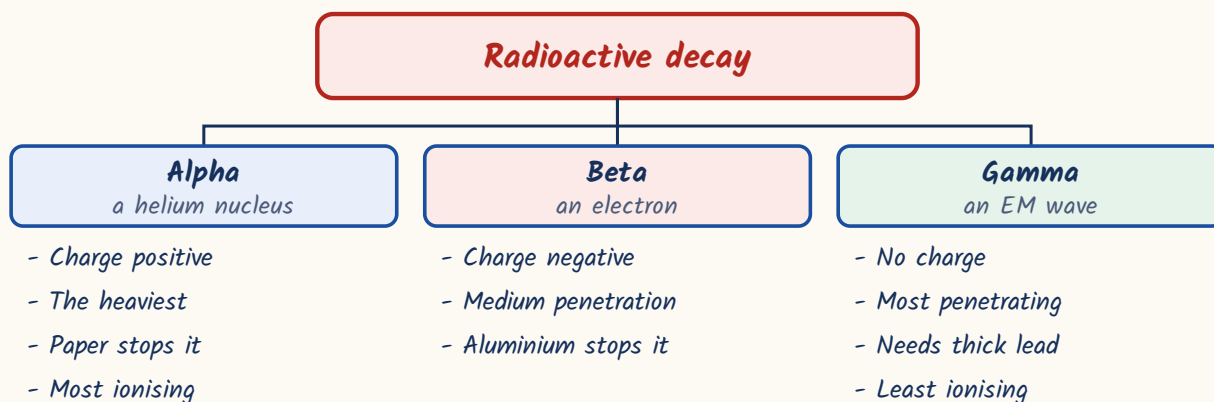
Asked in RAS Pre 2016 - a photovoltaic cell converts solar energy **DIRECTLY** into electrical energy, not into heat first. The fuse question - 'a thin wire of low melting point connected in series in the live wire' - is a documented repeat.

Modern physics – radioactivity, fission, fusion, and the master device table

Modern physics in RAS is a naming exercise: who discovered what, which radiation is which, and where fission ends and fusion begins. Radioactivity is the spontaneous breakdown of an unstable nucleus, and it throws out three kinds of thing – alpha, beta and gamma. Understand what each one actually is and the rest of the questions follow.

- Alpha particle = a helium nucleus, positively charged, heaviest, least penetrating but most ionising - stopped by a sheet of paper.
- Beta particle = an electron, negatively charged, medium penetration - stopped by a few millimetres of aluminium.
- Gamma ray = an electromagnetic wave, no charge, MOST PENETRATING - needs thick lead or concrete to stop it.
- Half-life is the time in which half of a radioactive sample decays; it is fixed for each isotope and unaffected by temperature or pressure.
- Discoveries to memorise: X-rays - W. C. Roentgen (first Physics Nobel, 1901); radioactivity - Henri Becquerel; the photoelectric effect explained by Albert Einstein (Nobel 1921); the neutron - James Chadwick.
- Superconductivity, in which resistance falls to exactly zero below a critical temperature and the material expels magnetic field (Meissner effect), was found by Kamerlingh Onnes in mercury in 1911.
- Moderator SLOWS neutrons - heavy water (D₂O) or graphite. Control rods ABSORB neutrons - cadmium or boron. Do not swap these two.
- India's Rawatbhata plant in Chittorgarh district, Rajasthan, uses pressurised heavy water reactors.
- The Sun's energy comes from the FUSION of hydrogen into helium - not from burning and not from fission.
- Radioisotope uses: cobalt-60 for cancer radiotherapy, iodine-131 for the thyroid, carbon-14 for dating fossils, sodium-24 for tracing blood flow and pipeline leaks.
- A particle accelerator speeds charged particles up to very high energy and smashes them together - the Large Hadron Collider at CERN is the best known, and it is where the Higgs boson was confirmed in 2012.

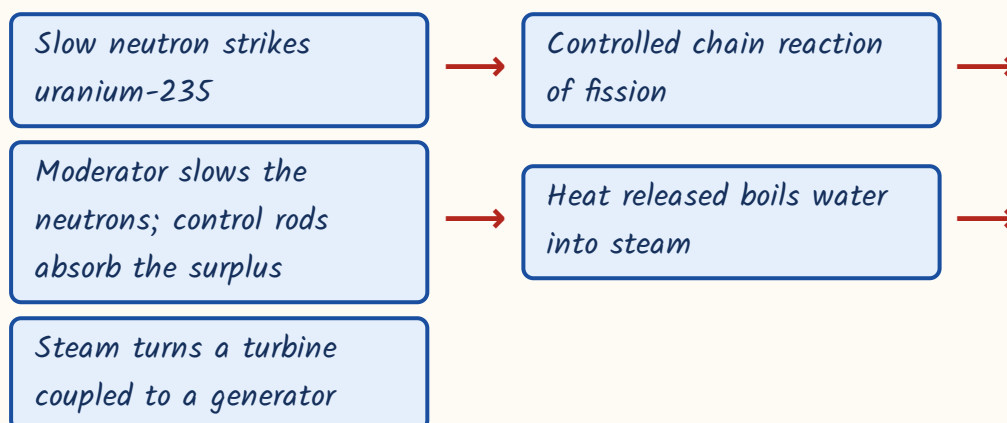
Alpha, beta and gamma compared



Nuclear fission vs nuclear fusion

Fission	Fusion
A heavy nucleus such as uranium-235 splits into lighter nuclei	Light nuclei such as hydrogen join to form a heavier one, helium
Triggered by a slow neutron; can be controlled	Needs enormous temperature and pressure; not yet controllable for power
Runs every nuclear power reactor, and the atom bomb	Powers the Sun and the stars, and the hydrogen bomb
Leaves highly radioactive waste	Produces far less radioactive waste

How a nuclear power station makes electricity



MASTER TABLE – device to principle. If you revise nothing else, revise this

Device or observation	Principle behind it
Hydraulic brake, hydraulic press	Pascal's law
Ship, submarine, lactometer, hydrometer	Archimedes' principle / law of flotation
Aeroplane lift, atomiser, chimney draught	Bernoulli's principle
Pressure cooker	Boiling point rises with pressure
Thermos flask	Blocks conduction, convection and radiation
Refrigerator, sweating, earthen pot	Cooling by evaporation / latent heat
Bimetallic strip in an iron or fire alarm	Unequal thermal expansion
SONAR, sonography, lithotripsy	Reflection of ultrasound
Stethoscope	Multiple reflection of sound
Mirage, diamond sparkle, optical fibre	Total internal reflection
Blue sky, red sunset	Scattering of light
Rainbow	Refraction + dispersion + internal reflection
Transformer	Mutual induction, AC only
Generator and electric motor	Electromagnetic induction; force on a current in a magnetic field

CURRENT LINK

Current link, as of September 2026: the 2025 Nobel Prize in Physics went to John Clarke, Michel Devoret and John Martinis for discovering macroscopic quantum-mechanical tunnelling and energy quantisation in an electric circuit - the physics that underlies superconducting quantum computers. Expect a straight 'who won it and for what' question; the phrase to hold on to is quantum tunnelling in a circuit.

CURRENT LINK

Current link, as of September 2026: India's National Quantum Mission, approved by the Union Cabinet in April 2023 with an outlay of about Rs 6,003 crore for eight years, is the scheme name to remember beside that Nobel. It aims at quantum computers of 50 to 1,000 physical qubits, satellite-based secure quantum communication and quantum sensors, and it works through four thematic hubs in IISc and the IITs. The United Nations also declared 2025 the International Year of Quantum Science and Technology, marking a hundred years of quantum mechanics - a ready-made 'which year was declared what' question.

YAAD RAKHO

Remember for the radiations: ALPHA is heavy and lazy, paper stops it. BETA is medium, aluminium stops it. GAMMA is the marathon runner, only lead stops it. Penetration order gamma > beta > alpha; ionising power is exactly the reverse.

REVISION RECAP

1. Seven SI base units - metre, kilogram, second, ampere, kelvin, mole, candela. Candela is the one students forget.
2. Light year and parsec are DISTANCE (1 parsec = 3.26 light years); diopetre is lens power; poise is viscosity; knot is speed; barn is area.
3. 1 atm = 76 cm of Hg = 1.013×10^5 Pa = 760 torr. 1 hp = 746 W. 1 kWh = 3.6×10^6 J.
4. First law = inertia (seat belt, carpet dusting). Third law = rocket, gun recoil, walking, swimming. Impulse explains the air bag.
5. g is maximum at the poles, minimum at the equator, zero at the Earth's centre, one-sixth on the Moon. Escape velocity 11.2 km/s.
6. Geostationary satellite - 36,000 km, equatorial, 24-hour period, for communication and weather. Polar orbit at 800-900 km for remote sensing.
7. Weightlessness in orbit is free fall, not the absence of gravity.
8. Pascal - hydraulic brake. Archimedes - ship, lactometer, hydrometer. Bernoulli - aerofoil, atomiser, two boats drawn together.

9. Pressure cooker cooks faster because higher pressure raises the boiling point; on a hill the reverse happens.
10. Water's specific heat 4.2 J/g/K is the highest among common liquids; latent heat of fusion 80 cal/g , of vaporisation 540 cal/g - hence steam burns worse.
11. Water is densest at 4°C , so ice floats and lakes freeze from the top downwards.
12. Conduction needs contact, convection needs a moving fluid, radiation needs nothing. A thermos defeats all three.
13. No sound in vacuum; fastest in solids; audible 20 Hz to 20 kHz ; echo needs 17.2 m ; stethoscope = multiple reflection.
14. Ultrasound - SONAR, sonography, echocardiography, lithotripsy, flaw detection. Doppler - train whistle, radar gun, red shift.
15. Total internal reflection - mirage, diamond, optical fibre. Scattering - blue sky, red sunset, black sky in space, red danger signals.
16. Myopia concave, hypermetropia convex, presbyopia bifocal, astigmatism cylindrical, cataract surgery. Convex power positive, concave negative.
17. EM spectrum by increasing wavelength: gamma, X-ray, UV, visible, infrared, microwave, radio. The TV remote uses infrared.
18. Fuse - low melting point wire, in series, in the live wire, replaced after blowing. MCB - trips, then is reset.
19. Transformer works on mutual induction and only on AC. Motor = Fleming's left hand; generator = Fleming's right hand.
20. Fission splits uranium-235 and runs reactors; fusion joins hydrogen and runs the Sun. Moderator slows neutrons, control rods absorb them. Gamma is the most penetrating radiation.